9 Video classification

- *video classification* represents the task of identifying actions in videos, as opposed to image classification, where only the most important object in the image has to be identified
- in video classification, the models have access to not only the appearance information present in single, static images, but also their complex temporal evolution

9 Video classification



Images: Recognize **objects**

Dog
Cat
Fish
Truck



Videos: Recognize **actions**

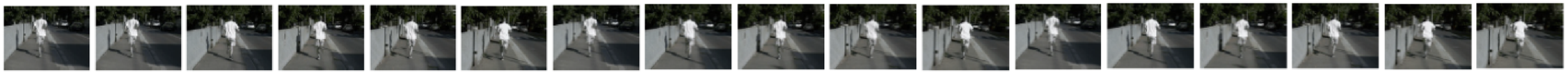
Swimming
Running
Jumping
Eating
Standing

9 Video classification

- the problem is that videos are big
- for example, an uncompressed video (3 bytes per pixel) in SD (640×480) resolution is ~ 1.5 GB per minute, and in HD (1920×1080) resolution is ~ 10 GB per minute
- the solution is to train on short clips, which have low fps and low spatial resolution, for example, $T = 16$, $H = W = 112$ (a 3.2 seconds video at 5 fps is just 588 KB)

9 Video classification

Raw video: Long, high FPS



Training: Train model to classify short **clips** with low FPS



Testing: Run model on different clips, average predictions

